iii) It is used as fibre - swelling agent in textile processing.

Evaluate Yourself

6) Haloalkanes undergo nucleophilic substitution reaction whereas haloarenes undergo electrophilic substitution reaction. comment.

14.6 Poly halogen compounds

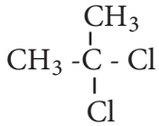
Carbon compounds containing more than one halogen atoms are called poly halogen compounds. Some of the important poly halogen compounds are described below.

They are classified as

a) gem – dihalides

CH_3CHCl_2

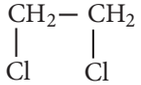
Ethylidene chloride
1,1 - Dichloro ethane



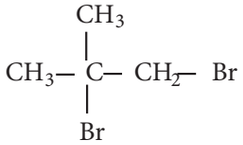
Isopropylidene chloride (or)
Isopropylidene dichloride
2,2-Dichloropropane

b) vic - dihalides

For Example



Ethylene dichloride (or) Ethylene chloride
1, 2 -Dichloro ethane



Isobutylene dibromide (or) Isobutylene bromide
1, 2-Dibromo-2-methyl propane

14.6.1 Preparation

a) gem- dihalides

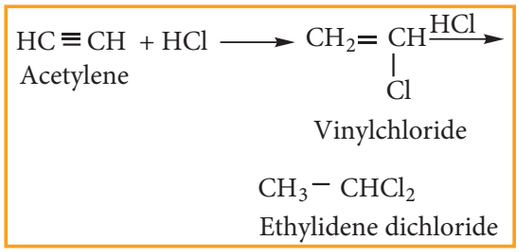
Ethylidene dichloride (1, 1 - Dichloro ethane) is prepared by

(i) Treating acetaldehyde with PCl_5



$CH_3CHO + PCl_5 \longrightarrow CH_3CHCl_2 + POCl_3$
Acetaldehyde — Ethylidene dichloride

(ii) Adding hydrogen chloride to acetylene

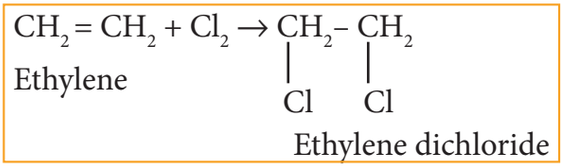


$HC \equiv CH + HCl \longrightarrow CH_2 = CH(Cl) \xrightarrow{HCl} CH_3 - CHCl_2$
Acetylene — Vinylchloride — Ethylidene dichloride

b) vic- dihalides

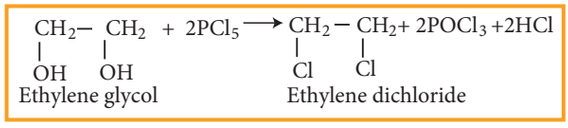
Ethylene dichloride (1, 2 - Dichloro ethane) is prepared by the following methods.

i) Addition of chlorine to ethylene



$CH_2 = CH_2 + Cl_2 \rightarrow CH_2(Cl) - CH_2(Cl)$
Ethylene — Ethylene dichloride

ii) Action of PCl_5 (or HCl) on ethylene glycol



$CH_2(OH) - CH_2(OH) + 2PCl_5 \longrightarrow CH_2(Cl) - CH_2(Cl) + 2POCl_3 + 2HCl$
Ethylene glycol — Ethylene dichloride

Properties

Physical Properties

i) They are sweet smelling, colourless liquids having relatively high boiling points.

Unit 14 - jagan.indd 245 22-01-2021 13:05:03